Лабораторная работа №3.1.Обыкновенные дифференциальные уравнения 1-го

:   . .
 1 5 3 5 0 3
  2 .

• Задание №1.

•   y y '   =   - 2 x ,
        ,   M (0 ,   5) .

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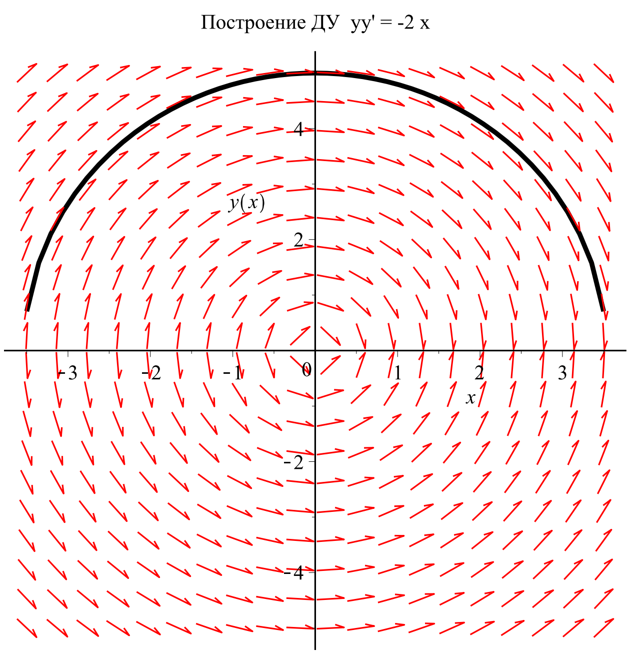
> with(DETools) :
DY1 := diff(y(x), x) = -2·x / y(x) :
DEplot(DY1, y(x), x =-3.5 ..3.5 , y(x) =-5 ..5, [y(0) = 5], linecolor = black, color = red,
    animatecurves = true, title = "Построение ДУ  yy' = −2 x ");
```

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> restart
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• Задание №2.

• **1)** , $M0(9,3)$, M MN Oy , $a=15$, Oy.

• **2)** , $M0(2,e)$, M MN Ox Ox, M. $a=$ -2.

2.1 :

y = $f(x0) + f'(x0)(x0 - x)$ - уравнение касательной к $f(x)$ в точке $x0$.

$y(t) = f(x) + \frac{1}{f'(x)}(t - x)$ - уравнение нормальной прямой.

(0 , $f(x) + \frac{1}{f(x)}$**) => y(0) =** $f(x) + \frac{x}{f'(x)}$.

$|MN| = a = \sqrt{x^2 + \frac{x^2}{f'(x)}}$

Так как MN образует острый угол с положительным направлением оси ординат, то $f(x) = \sqrt{a^2 - x^2} + C$

M0(9,3) => C = -9 => $f(x) = \sqrt{225 - x^2} - 9$.

> $y := simplify\left(\pm\int\frac{x}{\sqrt{(225-x^2)}}\,dx\right);$

$$y := \pm\text{-}\sqrt{-x^2+225} \quad (1)$$

M N Oy , y ' > 0 ⇒

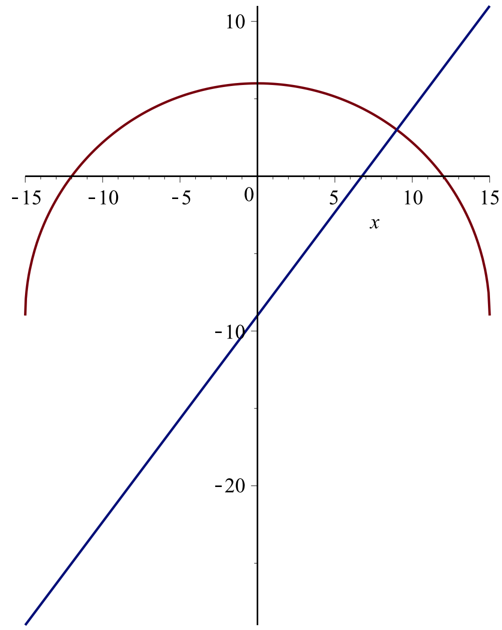
$y = \sqrt{225 - x^2} + C$ **=> [x = 9, y = 3] => C = -9.**

> $C := 3 - \sqrt{225 - 9^2};$
$f := \mathrm{sqrt}(225 - x^2) - 9;$

$k := subs\left(x=9, \frac{1}{diff(f,x)}\right):$
$n := subs(x=9, f) - k\cdot(x-9):$
$plot([f, n], \mathrm{x} = -15..15, \mathrm{scaling} = \mathrm{constrained});$

$$C := -9$$

$$f := \sqrt{-x^2+225} - 9$$



> *restart*

2 . 2 :

$\mathbf{y} = f(x0) + f\,'(x0)(x0 - x)$ *- уравнение касательной к* $f(x)$ *в точке* $x0$.

$Y = y,\ X = x,\ f(x0) = y.$

$Y - y = y'(X - x).$

По условию: $(x - xn) = \dfrac{a}{x}$

Точка, которая лежит на касательной, $N(xn, 0)$ *на ось абсцисс:* $y = y'(x - xn) \Rightarrow xn = x - \dfrac{y}{y'}$

$\Rightarrow y' = \dfrac{x \cdot y}{a}$

Найдем C *при* $[x = 2,\ y = e]$, $a = -2$;

$e = e^{-1 + C} \Rightarrow 1 = C - 1 \Rightarrow C = 2;$

$y = e^{2 - \frac{x^2}{4}}$;

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> dsolve(y' = x·y / a);
  C := 2;
```

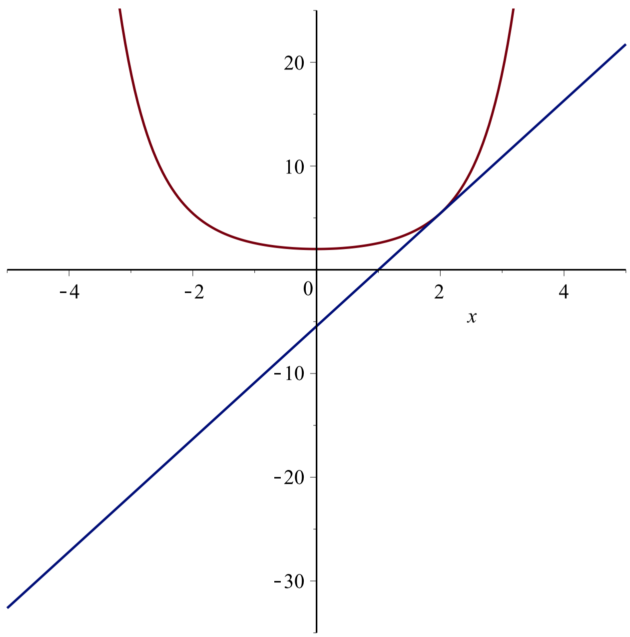
$$y(x) = \_C1\, \mathrm{e}^{\frac{1}{2} \frac{x^2}{a}}$$

$$C := 2 \qquad \textbf{(2)}$$

```
> a := -2 :
  DY := y(x)/diff(y(x), x) = -a/x;
  f := rhs(C · simplify(dsolve({DY, y(2) = e}, y(x))));
  k := (subs(x = 2, diff(f, x))) :
  m := subs(x = 2, f) + k·(x - 2) :
  plot([f, m], x = -5..5);
```

$$DY := \frac{y(x)}{\frac{\mathrm{d}}{\mathrm{d}x} y(x)} = \frac{2}{x}$$

$$f := 2\, \mathrm{e}^{\frac{1}{4} x^2}$$



> *restart*

• Задание №3.

•   .

-   .   .

(3)

>

 # Важно!!! У меня не получается решить ДУ в Maple, выдает результат как показано ниже, в тетради все решено!!!

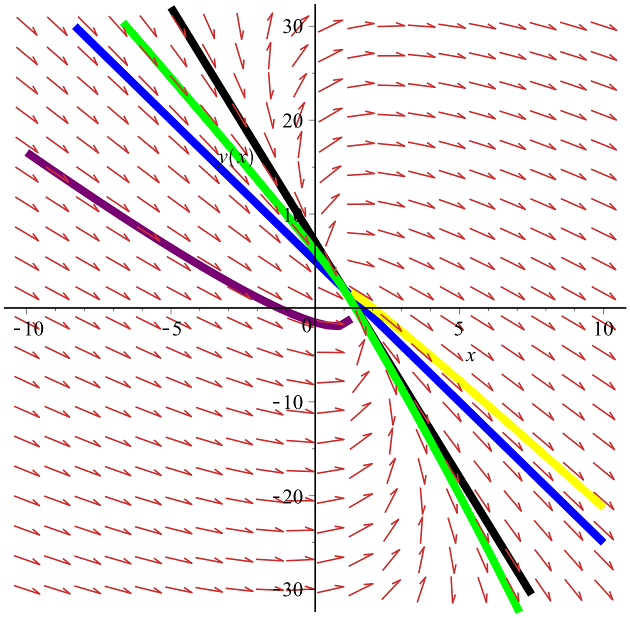
$dy := diff(y(x), x) = \dfrac{-15\cdot x + y(x) + 13}{9\cdot x + y(x) - 11};$

$dsolve(dy);$

$$dy := \frac{\mathrm{d}}{\mathrm{d}x}\, y(x) = \frac{-15\, x + y(x) + 13}{9\, x + y(x) - 11}$$

$$y(x) = 2 + \frac{1}{6}\,\frac{1}{\_C1}\left(6\sqrt{3}\,(x-1)\sqrt{\frac{(x-1)\,(27\,(x-1)\,\_C1-2)}{\_C1}}\,\_C1^2 + 54\,\_C1^2\,(x-1)^2 - 18\,(x-1)\,\_C1 + 1\right)^{1/3} - \frac{1}{6}\,(12\,(x-1)\,\_C1-1)\Bigg/\left(\_C1\left(6\sqrt{3}\,(x-1)\sqrt{\frac{(x-1)\,(27\,(x-1)\,\_C1-2)}{\_C1}}\,\_C1^2 + 54\,\_C1^2\,(x-1)^2 - 18\,(x-1)\,\_C1 + 1\right)^{1/3}\right) - \frac{1}{3}\,\frac{9\,(x-1)\,\_C1+1}{\_C1} - \frac{1}{2}\,\mathrm{I}\sqrt{3}\left(\frac{1}{3}\,\frac{1}{\_C1}\left(6\sqrt{3}\,(x-1)\sqrt{\frac{(x-1)\,(27\,(x-1)\,\_C1-2)}{\_C1}}\,\_C1^2 + 54\,\_C1^2\,(x-1)^2 - 18\,(x-1)\,\_C1 + 1\right)^{1/3} + \frac{1}{3}\,(12\,(x-1)\,\_C1-1)\Bigg/\left(\_C1\left(6\sqrt{3}\,(x-1)\sqrt{\frac{(x-1)\,(27\,(x-1)\,\_C1-2)}{\_C1}}\,\_C1^2 + 54\,\_C1^2\,(x-1)^2 - 18\,(x-1)\,\_C1 + 1\right)^{1/3}\right)\right) \quad \textbf{(4)}$$

> $with(DEtools) : DEplot\left(diff(y(x), x) = \frac{-15 \cdot x + y(x) + 13}{9 \cdot x + y(x) - 11}, y(x), x = -10..10, y(x) = -30..30, [y(2) = -1, y(2) = -3, y(2) = 0, y(2) = -3.2, y(-2) = 1.1], \text{color} = \text{orange}, animatecurves = true, linecolor = [blue, black, yellow, green, purple], thickness = 5\right);$



= [$\frac{a1}{a2}\ \frac{b1}{b2}$] -> {

$$\frac{a1 \cdot \text{alpha} + b1 \cdot \text{beta} + C1 = 0}{a2 \cdot \text{alpha} + b2 \cdot \text{beta} + C2 = 0}$$

$Det(A) = a1 \cdot b2 - b1 \cdot a1;$

Характеристическое уравнение $|A - \text{lambda} \cdot E| = 0 \Rightarrow \left[\frac{a1 - \text{lambda}}{a1}\ \ \frac{b1}{b2 - \text{lambda}} \right] \Rightarrow$

$\left\{ \frac{\text{lambda}1 = 6}{\text{lambda}1 = 4} \right. \Rightarrow$ ***Так как, два корня*** $\mathbf{> 0}$

$\Rightarrow$ ***особая точка является неустойчивым узлом.***

\>

$MatrixA := Matrix([[9, 1], [-15, 1]]);$

$CharacteristicEq := linalg[det]\big(MatrixA - \lambda \cdot Matrix([[1, 0], [0, 1]])\big) = 0;$

$RootsLambda := solve(eq, \lambda);$

$$MatrixA := \begin{bmatrix} 9 & 1 \\ -15 & 1 \end{bmatrix}$$

$$CharacteristicEq := \lambda^2 - 10\,\lambda + 24 = 0$$
$$RootsLambda := 6, 4 \qquad \textbf{(5)}$$

> *restart*

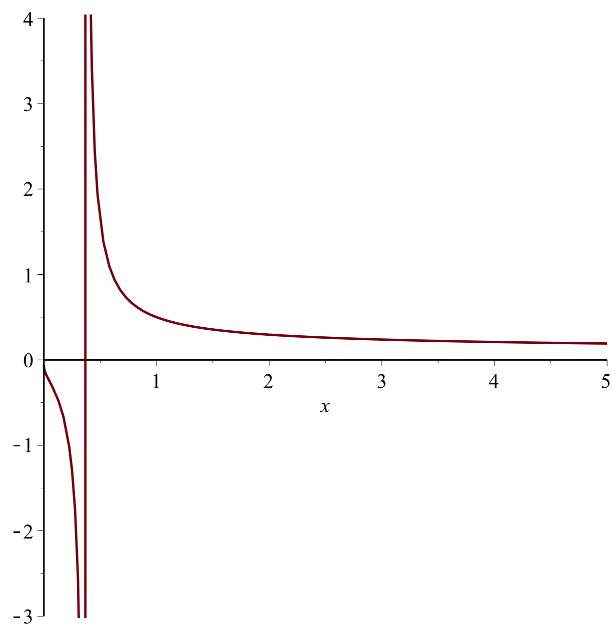
• Задание №4.

• . .

> $DY := simplify\left(\left(dsolve\left(\left\{x \cdot y' + y = 2 \cdot \ln(x) \cdot y^2, y(1) = \frac{1}{2}\right\}\right)\right)\right);$
$DY := rhs(\%);$

$plot(DY, x = -5 .. 5);$

$$DY := y(x) = \frac{1}{2\ln(x) + 2}$$
$$DY := \frac{1}{2\ln(x) + 2}$$



> *restart*

- **Задание №5.**
- .

-1 1 .

5.1 x = 2 * y' * arctan(y') - ln(y'2 + 1)

> $X := 2 \cdot t \cdot \arctan(t) - \ln(t^2+1);$

$$X := 2\, t \arctan(t) - \ln(t^2+1) \tag{6}$$

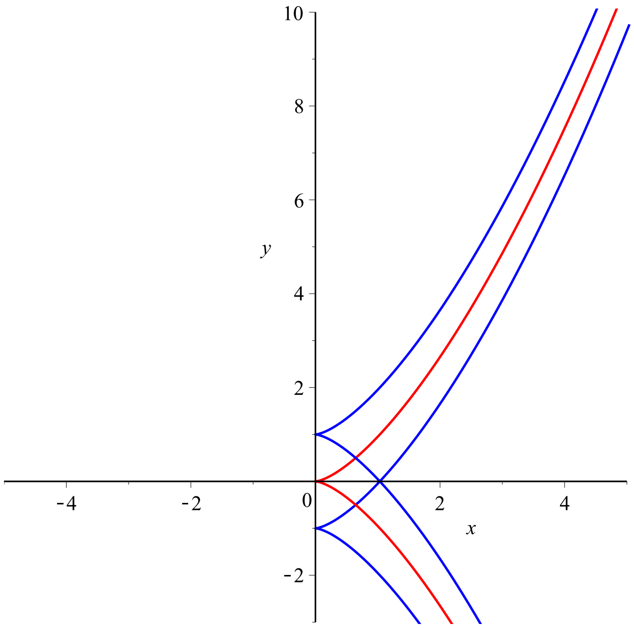
> $dx := diff(X, t);$

$$dx := 2 \arctan(t) \tag{7}$$

> $Y := dsolve(diff(y(t), t) = dx \cdot t);$

$$Y := y(t) = \arctan(t)\, t^2 - t + \arctan(t) + \_C1 \tag{8}$$

> $plot1 := plot\left(\left[X, \frac{1}{2} \cdot dx \cdot t^2 - 1, t = -5..5\right], x = -5..5, y = -3..10, color = blue\right):$
$plot2 := plot\left(\left[X, \frac{1}{2} \cdot dx \cdot t^2, t = -5..5\right], x = -5..5, y = -3..10, color = red\right):$
$plot3 := plot\left(\left[X, \frac{1}{2} \cdot dx \cdot t^2 + 1, t = -5..5\right], x = -5..5, y = -3..10, color = blue\right):$
$plots[display](plot1, plot2, plot3);$



> *restart*

5.2 y = y'chy' - shy'

> $Y := t\cdot\cosh(t)\ -\ \sinh(t);$

$$Y := t\cosh(t) - \sinh(t) \tag{9}$$

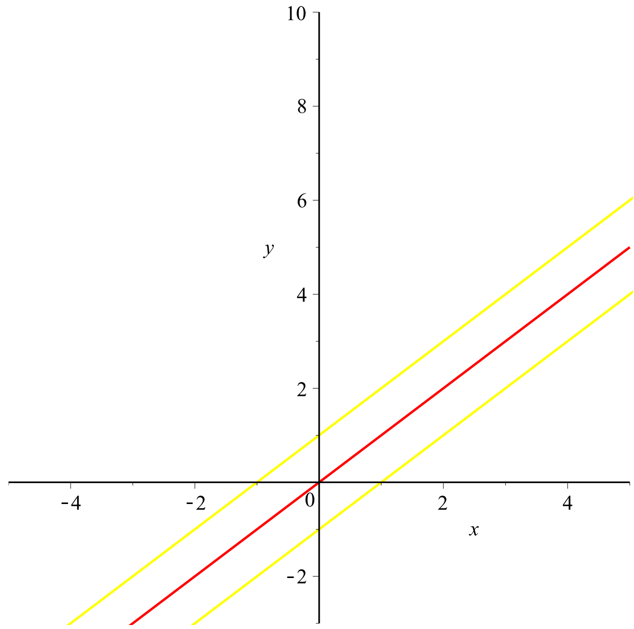
> $dy := diff(Y, t);$

$$dy := t\sinh(t) \tag{10}$$

> $X := dsolve(diff(x(t), t) = t\cdot\sinh(t), x(t));$
> $X := rhs(\%);$

$$X := x(t) = t\cosh(t) - \sinh(t) + \_C1$$
$$X := t\cosh(t) - \sinh(t) + \_C1 \tag{11}$$

> $plot1 := plot([t\cosh(t) - \sinh(t) + 1, \mathrm{Y}, t = -5..5], x = -5..5, y = -3..10, color = yellow):$
> $plot2 := plot([t\cosh(t) - \sinh(t), \mathrm{Y}, t = -5..5], x = -5..5, y = -3..10, color = red):$
> $plot3 := plot([t\cosh(t) - \sinh(t) - 1, \mathrm{Y}, t = -5..5], x = -5..5, y = -3..10, color = yellow):$
> $plots[display](plot1, plot2, plot3);$



> *restart*

• Задание №6.

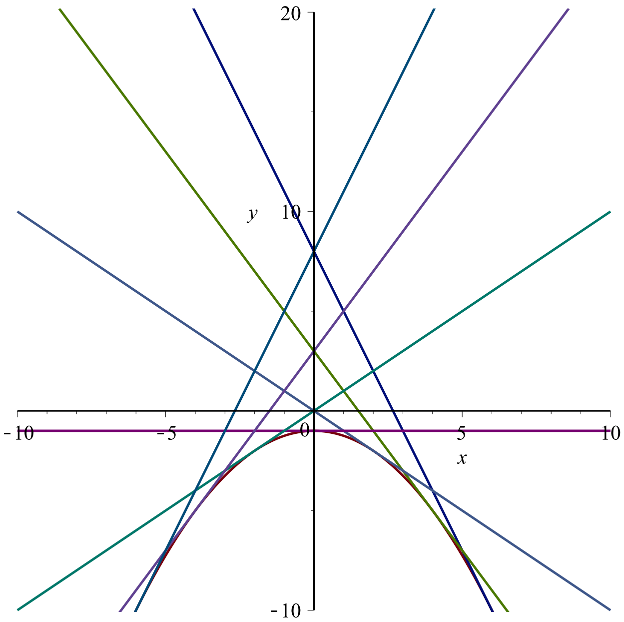
• .

 - 3 3 .

> $DY := dsolve\left(y = x \cdot y' + y'^2 - 1\right)$

$$DY := y(x) = -\frac{1}{4}\,x^2 - 1,\, y(x) = \_C1^2 + \_C1\,x - 1 \qquad \textbf{(12)}$$

> $plot([rhs(DY[1]), seq(subs(\_C1 = i, rhs(DY[2])), i = -3\,..3)], x = -10\,..10\,, y = -10\,..20);$



20
10
y
-10
-5
0
5
10
x
-10

>